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[22CSE303]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

NOSQL DATABASE USING MONGODB

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Can you summarize the main features of MongoDB?	L1	1	**02**
	b.	Write a MongoDB query to retrieve a limited number of documents from a collection.	L3	2	**02**
	c.	Identify the purpose of indexing in MongoDB.	L1	3	**02**
	d.	Define the purpose of an arbitrator in replication.	L2	4	**02**
	e.	Identify two differences between MongoDB and a relational database.	L3	5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Describe the different types of data modeling strategies in MongoDB and provide examples of their use.	L3	1	**05**
		OR			
	b.	Classify different types of NoSQL databases (CP, AP, CA) according to the CAP theorem, and give examples.	L2	1	**05**
3.	**a.**	Compare and contrast document-based databases with key-value databases.	L2	1	**05**
		OR			
	b.	Demonstrate with examples how to apply limiting and filtering to retrieve specific sets of documents in MongoDB.	L2	2	**05**
4.	**a.**	Apply the 'Filtering' documents query in MongoDB with suitable example.	L3	2	**05**
		OR			
	b.	How would you use MongoDB functions to limit the number of documents returned in a query? Provide examples.	L3	2	**05**
5.	**a.**	Differentiate $limit and $skip functions with suitable examples.	L4	3	**05**
		OR			
	b.	Analyze the purpose of different aggregation commands in MongoDB. How do they enhance data querying and transformation?	L4	3	**05**
6.	**a.**	Analyze the role of replication in MongoDB. How does it impact data consistency, availability, and fault tolerance?	L4	4	**05**

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		OR			
	b.	Define arbitrator in MongoDB replication.	L2	4	**05**
7.	**a.**	Explain the concept of indexing a shard cluster.	L2	5	**05**
		OR			
	b.	List the various steps involved in Deployment of cluster.	L1	5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Explain MongoDB architecture in detail with proper diagram.	L2	1	**10**
		OR			
	b.	Analyze the different types of NoSQL databases in detail.	L4	1	**10**
9.	**a.**	Compare MongoDB with a traditional relational database management system (RDBMS) across various parameters.	L3	1	**10**
		OR			
	b.	Explain in detail the process of creating, updating, deleting, and querying documents in MongoDB.	L2	2	**10**
10.	**a.**	Demonstrate CRUD operations for any sample collection with five documents.	L2	2	**10**
		OR			
	b.	Discuss how filtering and limiting documents are achieved through MongoDB queries.	L3	2	**10**
11.	**a.**	Examine the need of Indexing in MongoDB with suitable example.	L4	3	**10**
		OR			
	b.	Solve the various String functions in MongoDB with proper example	L3	3	**10**
12.	**a.**	Outline the process of implementing replication in MongoDB with an example setup.	L2	4	**10**
		OR			
	b.	Explain the concept of replication in MongoDB and discuss its advantages.	L3	4	**10**
13.	**a.**	Evaluate the process of setting up a sample shard cluster in MongoDB. What are the challenges and best practices for building a scalable, fault-tolerant system?	L5	5	**10**
		OR			
	b.	Analyze the components of a MongoDB sharded cluster. How do they contribute to horizontal scaling and data distribution across multiple servers?	L4	5	**10**